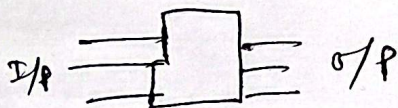


* Difference betⁿ Combination Circuit and Sequential

Combⁿ Ckt

① o/p is only dependent on present I/P



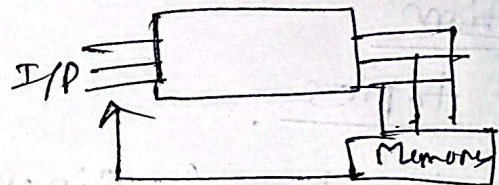
- No feedback
- No memory

Eg Half Adder / FA

- Half subtractor / FS
- Mux / Demux
- Encoder / Decoder

Seqⁿ Ckt

→ o/p is depending on present I/P and o/p



- feedback
- memory

Eg Counter

- shift register
- Flip Flop

MUX Tree design :-

- mux tree is used to obtain higher order mux using lower order mux

Q- How many 4×1 mux is req^d to make 32×1 mux

$$\frac{32}{4} = 8$$

$$= 8 + 2 + 1$$

$$\frac{8}{4} = 2$$

$$= 11$$

Ans.

$$\frac{2}{4} = 0.5 < 1$$

Q2 Design of 4×1 mux using 2×1 mux

Ans. Requirement

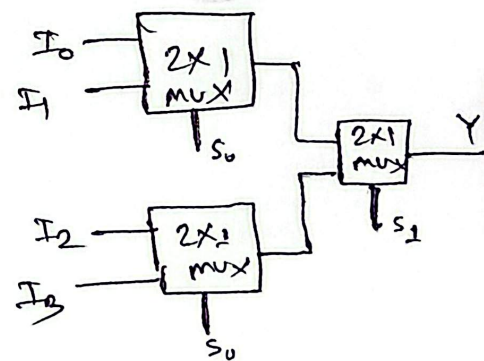
$$\frac{4}{2} = 2$$

$$2 + 1 = 3$$

$$\frac{2}{2} = 1$$



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



* Designing 8x1 mux using 2x1 mux

Ans:-

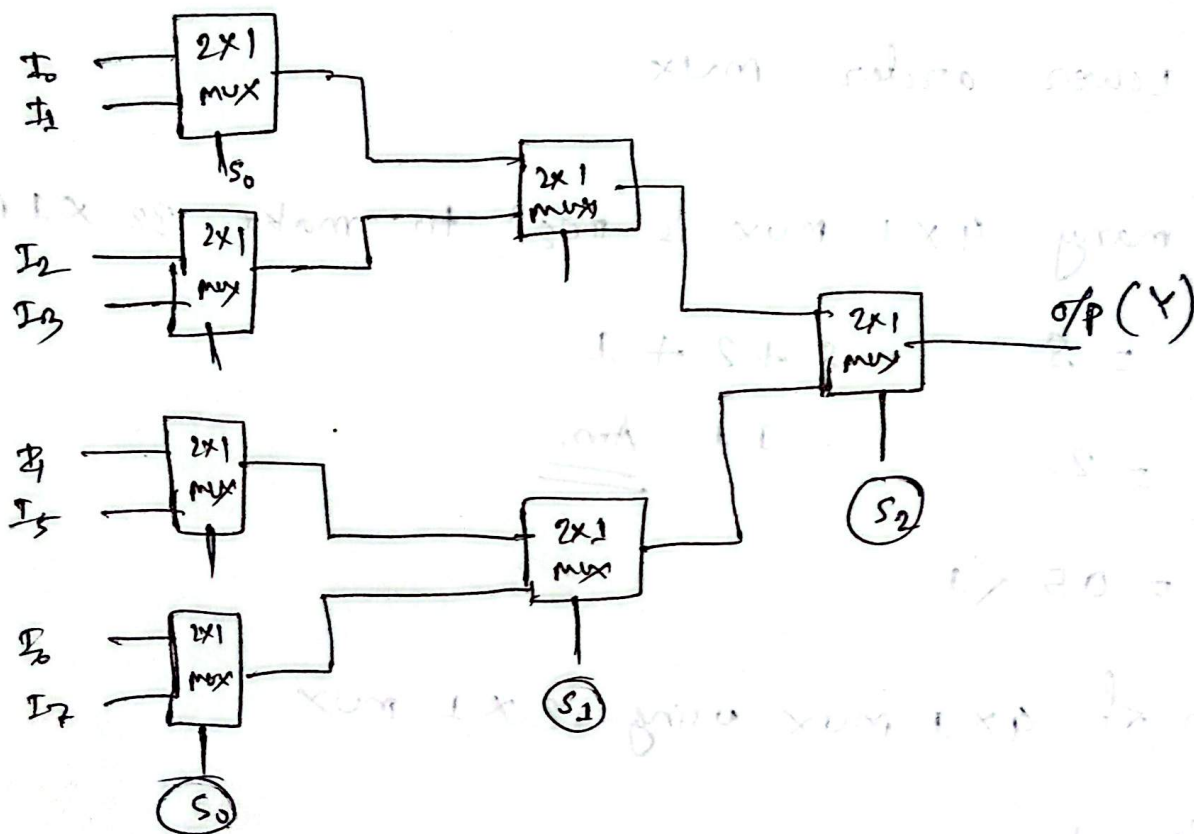
$$\frac{8}{2} = 4$$

$$\frac{4}{2} = 2$$

$$\frac{2}{2} = 1$$

$$4 + 2 + 1 = 7$$

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7



SOP Implementation by mux

Function $F(A, B, C, D) = \sum m(0, 2, 6, 9, 11, 13)$ make it by 4x1 mux

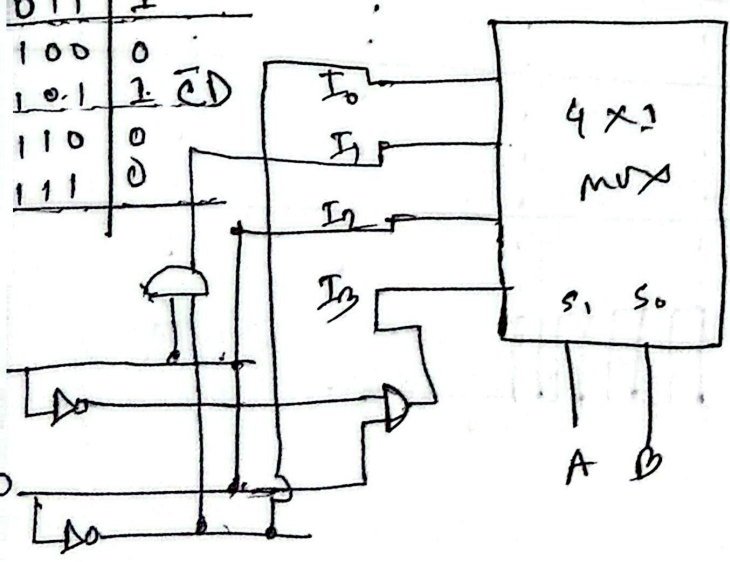
Ans

BCD	F
000	1
001	0
010	1
011	0
100	0
101	0
110	1
111	0

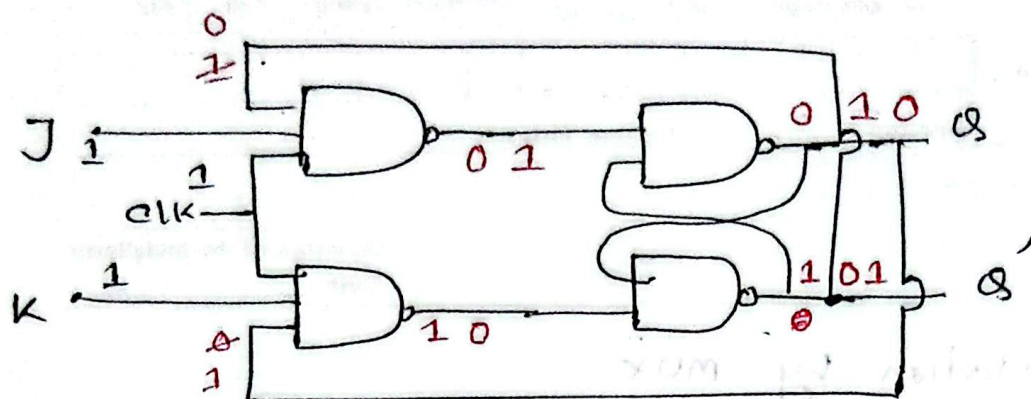
AB \ CD	00	01	11	10
00	1			1
01				1
11		1		
10		1	1	

s_1	s_0	y
00		I_0
01		I_1
10		I_2
11		I_3

AB	
00	$\rightarrow D$
01	$\rightarrow \bar{C}$
10	$\rightarrow D$
11	$\rightarrow \bar{C}D$



* Race Around Conditions in JK FF



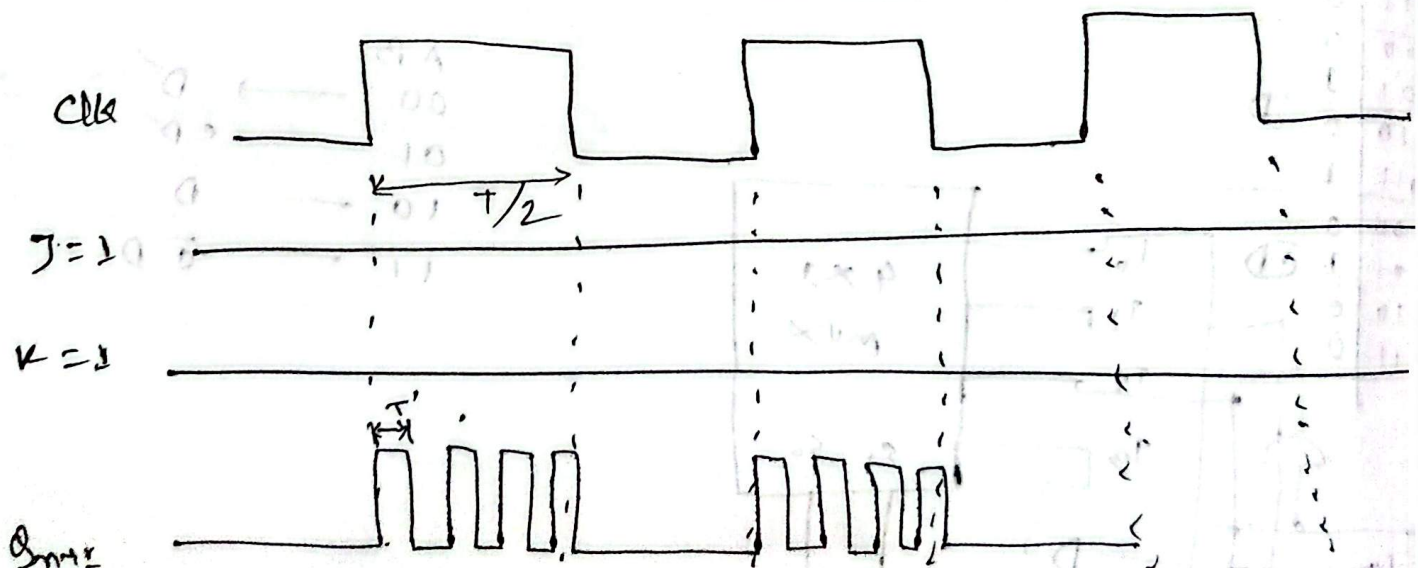
Toggleing

Truth table of JK ff

clk	J	K	Q	Q'
0	X	X	memory	
1	0	0	memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	memory	

continuous change of
(Toggleing until the clock
changed) This is called
race around condition.

race around condition



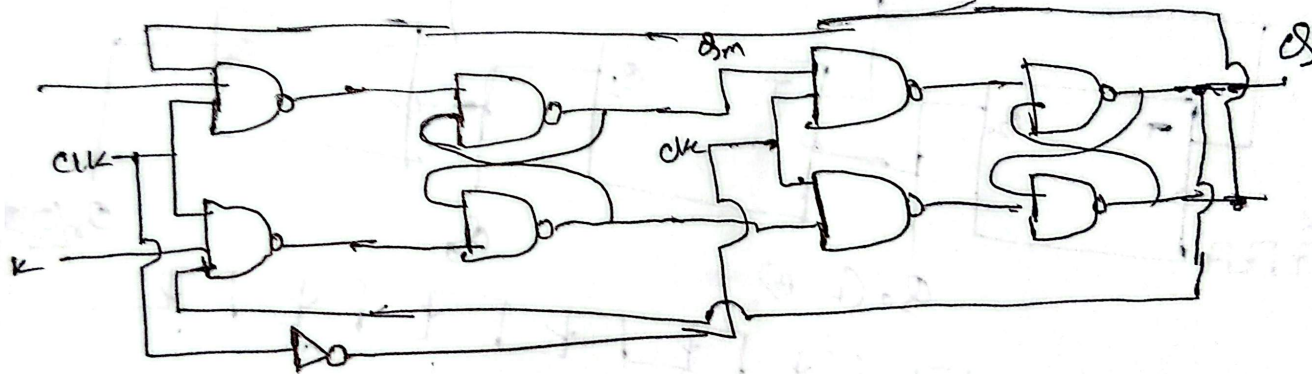
Solⁿ 1) $T' > T/2$ propagation delay greater than $T/2$

2. Edge Triggered

3. Master Slave Flip Flop

(provides -ve Edge Trigger)

Master Slave Flip-Flop

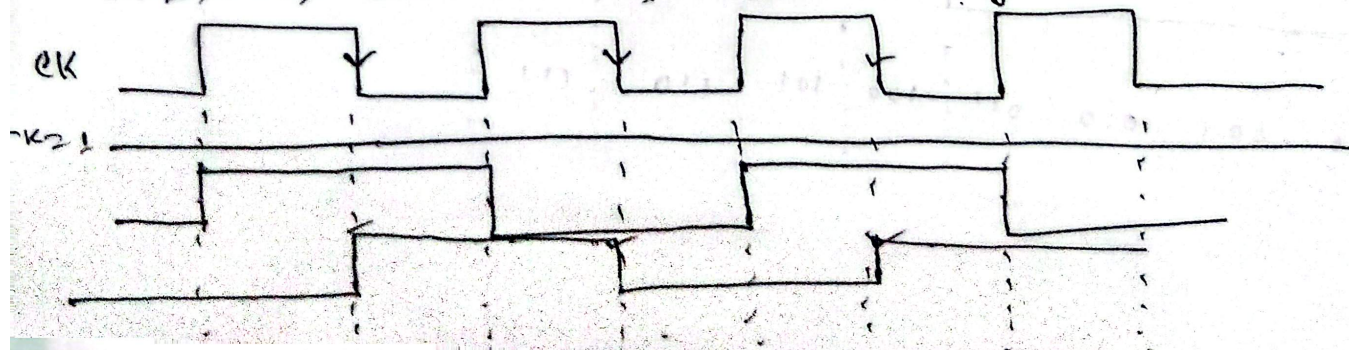


Master

Slave

$J=1, K=1, clk=1, Q_m = \text{Active}, Q = \text{Memory}$

$J=1, K=1, clk=0, Q_m = \text{Memory}, Q = \text{Active}$



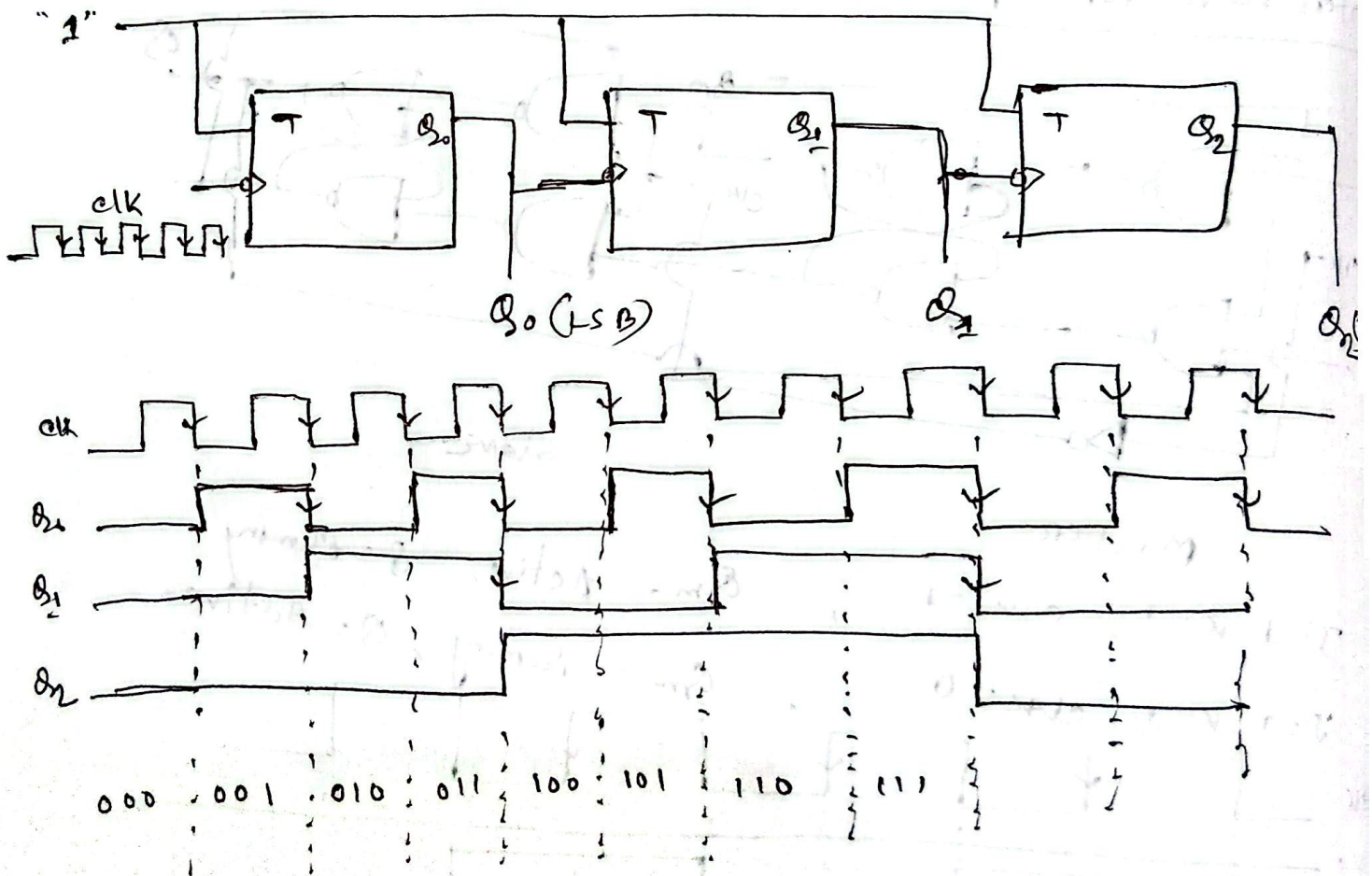
Counter

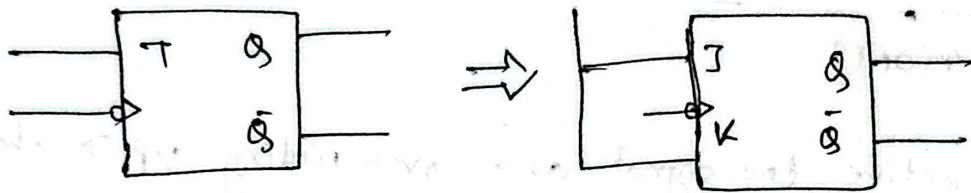
- Up counter $[0, 1, 2, \dots, N]$
- Down counter $[N, N-1, \dots, 1, 0]$

classification

- 1) Synchronous counter (memory elements have same clock)
 - 2) Asynchronous counter (memory elements have different clock pulse)
- For n bits counter, max count $= 2^n$

Ripple Counter (Asynchronous Counter)





Modulo Counter :-

mod 6 counter (0 to 5 counts)

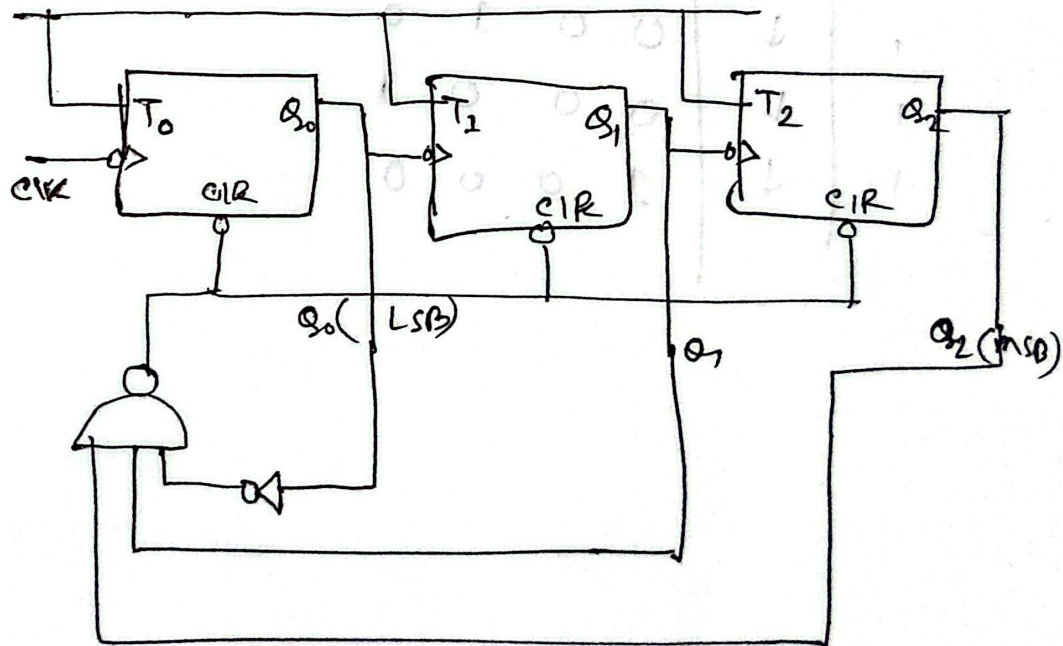
bits 3 ($2^3 = 8$)

mod 12 " (0 to 11 counts)

bits 4 ($2^4 = 16$)

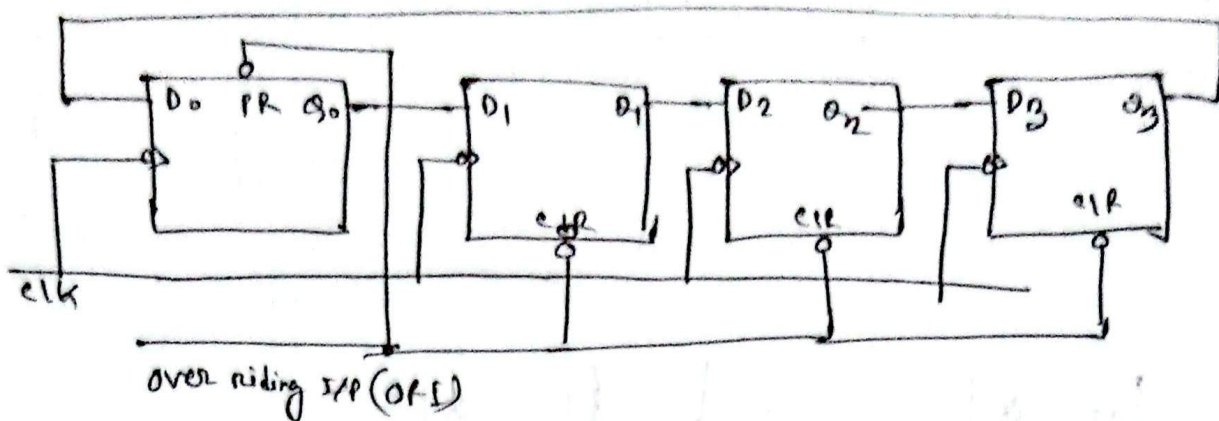
Mod 6:

Q_2	Q_1	Q_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1



Ring Counter

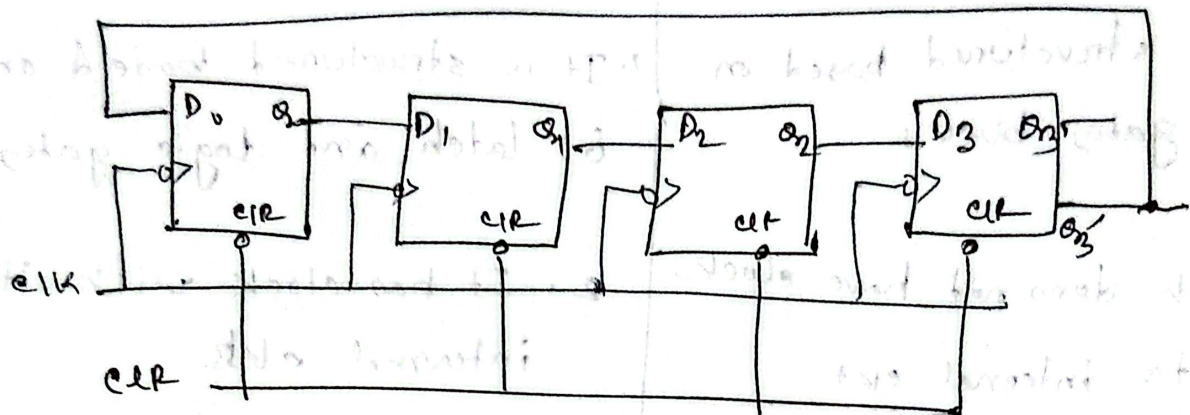
→ It is an application of shift register



when giving active low signal as an over riding i/p to the PR
no matter the i/p Q_0 will be 1, and clk makes $Q_1=0$
 $Q_2=0$, $Q_3=0$

OR.I	clk	Q_0	Q_1	Q_2	Q_3
	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

Johnson's Counter:-

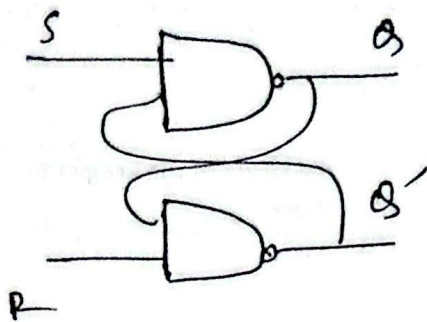


clr	clk	Q ₀	Q ₁	Q ₂	Q ₃
	X	0	0	0	0
↓	↓	1	0	0	0
↓	↓	1	1	0	0
↓	↓	1	1	1	0
↓	↓	1	1	1	1
↓	↓	0	1	1	1
↓	↓	0	0	1	1
↓	↓	0	0	0	1
↓	↓	0	0	0	0

$$Q_3' = 1 \rightarrow D = 1 \rightarrow Q_0 = 1$$

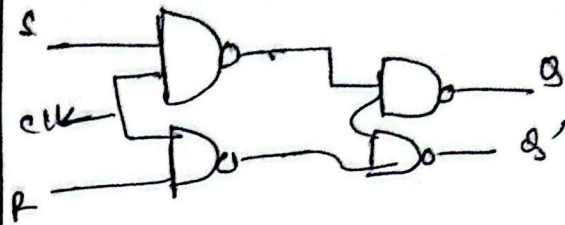
Latch vs FlipFlop

Latch



1. It is structured based on logic gates blocks
2. Latch does not have clock in its internal ckt
3. Change output corresponding to changing the i/p only
4. It is level trigger
5. Requires less power
6. Faster

FlipFlop



2. It is structured based on block & latch and logic gates
2. It has clock with its internal ckt.
3. It change the o/p w.r. to i/p and clock signal
4. It is edge trigger
5. Requires more power
6. Slower than latch